Md Zesun Ahmed Mia

Student Researcher | Machine Learning for ASIC Design & AI-Driven EDA

PhD Candidate at Penn State specializing in ML-hardware co-design and AI-driven IC optimization. Researching robust neural architectures and accelerator designs that mitigate circuit non-idealities through bio-inspired plasticity and advanced cross-layer simulations for high-performance computing.

Education

08/2022–05/2027 (Expected) **Ph.D. in Electrical Engineering**, *Pennsylvania State University*, University Park, PA, USA, *CGPA: 4.00/4.00*
Advisor: Dr. Abhronil Sengupta

08/2022–08/2024 **M.S. in Electrical Engineering**, *Pennsylvania State University*, University Park, PA, USA, *CGPA: 4.00/4.00* | Thesis: Neuromorphic Computing for Lifelong Learning

02/2015–04/2019 **B.Sc. in Electrical and Electronic Engineering**, *Bangladesh University of Engineering & Technology (BUET)*, Dhaka, Bangladesh, *CGPA: 3.81/4.00*

Academic Research and Teaching Experience

08/2022–Present **Graduate Research Assistant**, *Penn State*, University Park, PA
- Conducted research on AI/ML methods for accelerating IC design; developed robust algorithms for specialized AI hardware; integrated ML-driven design flows with industry-standard CAD tools for RTL-level PPA evaluation.
- Directed characterization of hardware-algorithm interactions; built compact models in Verilog-A and cross-layer simulation frameworks using Python and MATLAB for specialized accelerators.
- Designed bit-serial accelerators for Transformer workloads; implemented CIM architectures with custom dataflow optimization and non-volatile memory integration.

08/2024–05/2025 **Graduate Teaching Assistant**, *Penn State*, State College, PA
- Taught Cadence Virtuoso (schematic/layout), PDK usage, DRC/LVS, and analog/digital design flows; created hands-on lab modules and guided tool/debug workflows.
- Supervised Capstone projects for 90+ undergraduate students across communications, electronics, and firmware: supported Raspberry Pi/Arduino development, software-defined radio experiments, PCB design, and system integration end-to-end.

02/2021–08/2022 **Lecturer**, *ULAB*, Dhaka, Bangladesh
Taught Solid State Devices, Digital Circuit Design, and Semiconductor Device Physics.

Technical Skills

Digital/ASIC	Logic synthesis, Digital simulation (Gate/transistor-level), Processor architecture, RTL design, RTL PPA evaluation, Verilog/SystemVerilog, FPGA prototyping, ASIC design flow, Memory/SRAM design, Computer architecture
ML Systems & Accel.	AI/ML-driven IC design algorithms; Robustness to hardware non-idealities; Circuit noise/mismatch aware ML; AI accelerator research; inference engine design; Systolic Arrays; Transformer/LLM; model compression (PTQ/QAT, pruning, distillation); dataflow optimization; hardware-ML co-design; latency-energy benchmarking
Hardware & EDA	VLSI design automation, Cadence Virtuoso, Spectre, HSPICE, TCAD, ModelSim, Vivado, CMOS/FinFET, Memory array design, Circuit characterization, Spintronic/Neuromorphic characterization
Programming	Python, CUDA, C/C++, Linux CLI, Bash/Shell, Docker, performance modeling, architectural simulation, parallel computing
AI Frameworks	PyTorch, TensorFlow, JAX, ONNX, TensorRT; model development and debugging; inference performance tuning; data tooling (Pandas, NumPy, Jupyter), experiment tracking (W&B, Matplotlib)
DTCO & STCO	Design-technology and system-technology co-optimization, cross-layer PPA, process-aware design, system-architecture codesign
Device & Process	Spintronic/Neuromorphic hardware, FeFETs, advanced memory, process integration, TCAD modeling, device-circuit co-design, characterization (DC/AC, IV/CV, AFM, SEM), metrology

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AI Dev.	Generative and agentic AI tools (Cursor, Copilot, etc.) for research, design, and code development
Collaboration	Git, Research Leadership, Technical Communication, Project Management

Industrial Experience

05/2025–07/2025 **Graduate Technical Intern**, *Intel Corporation*, Hillsboro, OR

- Designed comprehensive Design of Experiments (DOE) for thin film deposition optimization supporting high-volume semiconductor manufacturing.
- Implemented AI-powered models for automated recipe optimization and real-time monitoring in manufacturing environment.
- Developed statistical analysis frameworks to assess variability and establish robust control limits for production scalability.

09/2020–07/2021 **R&D Engineer**, *SEMWAVES Ltd.*, London, UK, part-time

- Delivered a 50 kW hybrid renewable system (solar + smallscale hydro) for an offgrid Bangladeshi site, supporting reliable community power.
- Owned technical leadership and coordination (vendors, field teams, stakeholders); directed device selection, integration, QA/safety procedures, and optimization against load profiles and uptime targets.

Projects

ML-Driven IC Design Acceleration	Lead Student Researcher, 2023–Present. Developed AI/ML methods to accelerate IC design via automated PPA evaluation and cross-layer simulation. Participated in the development of AI/ML-driven design algorithms robust to hardware mismatch and failures in CIM platforms.
Astromorphic Transformer	Lead Student Researcher, 2022–2025. Developed a bioplausible transformer architecture robust to circuit non-idealities through bio-inspired plasticity. Achieved 33.8 perplexity on WikiText-2 and improved convergence on image classification. See: [IEEE TCDS 2025].
MIPS Microprocessor Design	Lead Student Researcher, 2018–2019. Designed a 5-stage pipelined processor in Verilog; verified functionality through simulation and synthesized for FPGA. Demonstrated core knowledge of processor architectures and digital design flows.
Neuromorphic Cybersecurity	Lead Student Researcher, 2023–2025. Developed Hierarchical D-SNN for adaptive intrusion detection; achieved 94.3% accuracy. Focused on predicting system behaviors and adapting to dynamic conditions with minimal supervision. See: [ICONS 2025].

Publications

[1] **Md Zesun Ahmed Mia** et al. "Delving deeper into astromorphic transformers". In: *IEEE TCDS* (2025).

[2] **Md Zesun Ahmed Mia** et al. "Neuromorphic Cybersecurity with Semi-Supervised Lifelong Learning". In: *Proc. ICONS*. 2025.

[3] Arnob Saha, **Md Zesun Ahmed Mia**, et al. "Toward Variation-Tolerant Ferroelectric Neural Computing". In: *IEEE MWSCAS*. 2025.

[4] **Md Zesun Ahmed Mia** et al. "Impact of Doping and Defects on Thermal Transport of Monolayer GaN Nanoribbons". In: *IEEE ICECE*. 2024.

[5] T. Zhang et al. "Self-sensitizable neuromorphic device based on adaptive hydrogen gradient". In: *Matter* (2024).

[6] K.M.A.H. Fahim et al. "Study of 3-nm Cylindrical GAAFETs with Variations in High-k Dielectric". In: *IEEE ISIEA*. 2022.

Recognitions

- The Wormley Family Graduate Fellowship, Harry G. Miller Fellowships in Engineering (2025)
- Arthur Waynick Graduate Scholarship (2024),Milton and Albertha Langdon Memorial Fellowship (2023), Melvin P. Bloom Memorial Fellowship (2022)

Professional Affiliations

- Reviewer, IEEE TNNLS (2025), Design Automation Conference (DAC) (2025), IEEE MWSCAS (2025)
- Student Member, IEEE (2015-Present), Executive Member, EDS, IEEE BD (2021-2022)

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